

Retail Sales Analysis

Project Overview

This project was created to practice SQL and relational database concepts using PostgreSQL. The database simulates a small retail clothing store and includes information about products, customers, and sales transactions.

The project covers database creation, data insertion, filtering, sorting, aggregation, and business analysis queries using SQL.

Tools Used

- PostgreSQL
- PgAdmin 4
- DBeaver
- SQL

Database Structure

The database contains three related tables:

Products

Stores product information such as:

- product name
- category
- price
- stock quantity

Customers

Stores customer details including:

- customer name
- email
- phone number
- address

Sales

Stores sales transactions and connects customers with purchased products through foreign keys.

SQL Concepts Practiced

- CREATE DATABASE
- CREATE TABLE
- INSERT INTO
- UPDATE
- DELETE
- SELECT
- WHERE
- ORDER BY
- LIKE
- BETWEEN
- GROUP BY
- Aggregate Functions
- JOIN Operations
- Foreign Keys

Business Analysis Queries

The project also includes SQL queries focused on business insights, such as:

- Top selling products
- Revenue by product category
- Customer purchase history
- Product count by category

These queries were created to simulate real-world retail analysis scenarios.

Files Included

- SQL project file
- Database screenshots
- Query analysis screenshots

Notes

All SQL queries were executed using PostgreSQL.

PgAdmin 4 was used for database creation and management, while DBeaver was used for querying, analysis, and project documentation screenshots.